

# ASWIN K J

## AIML Undergraduate

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### SUMMARY

AIML undergraduate specialising in LLMs and RAG-based agent systems, with hands-on experience designing and deploying production-ready AI apps; Azure and Oracle Cloud Certified. Grounded understanding of data structures and algorithms in Python, backed by wins at 2+ hackathons. Experienced in leading technical teams and organising large-scale events, coordinating 10+ member teams / 1000+ attendee events. Seeking an AI Engineering internship to build scalable, production-grade AI systems.

### WORK EXPERIENCE

#### ML Intern | FlyRank | Remote | July 2026 - Present - [Offer Letter](#)

- Engineered semantic clustering and intent-classification pipelines using embeddings and unsupervised learning (e.g., HDBSCAN, scikit-learn) to map search-query demand to content-coverage gaps.
- Developed opportunity-scoring models translating raw search/analytics data into ranked, actionable recommendations, connecting model output directly to business-impact decisions.

#### DA Intern | LabMentix | Remote | July 2025 - Dec 2025 - [Completion Letter](#)

- Engineered deep learning pipelines using TensorFlow and EfficientNetB0 transfer learning, integrating MLflow for rigorous MLOps experiment tracking and hyperparameter versioning.
- Architected and deployed real-time predictive analytics inference dashboards using Streamlit, seamlessly bridging backend machine learning logic with interactive web applications.

### EDUCATION

#### B. TECH in ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

CGPA : 9.27/10 ( **Department 1st Rank** ) | SNS College of Technology | Coimbatore, India | Present

#### HIGHER EDUCATION | X<sup>th</sup> | XII<sup>th</sup>

School Head Boy | The Oxford School | Trivandrum, India | March 2022 - March 2024

### PROJECTS

- V2C : Voice to Code AI Assistant | 1st Place, Hack Hive 2025 - [GitHub](#)**
  - Architected a voice-to-code translation Flask backend with the Gemini API, optimising natural language parsing for low-latency, multi-language code generation.
  - Engineered a zero-click, voice-driven development state machine integrating the Web Speech API with custom audio heuristics for visually impaired accessibility.
  - Automated repository provisioning and version control pipelines using the GitHub REST API for dynamic multi-file packaging
- Exoplanet AI | NASA Space Apps Challenge - [GitHub](#)**
  - Engineered a 5-model deep learning ensemble (CNN, LSTM, Bi-Attention, RF, Hybrid) achieving 94.1% accuracy and 97.8% AUC on NASA Kepler/TESS photometry data.
  - Built a Flask inference backend with a time-series preprocessing pipeline, delivering sub-200ms predictions using 20+ extracted spectral features.
  - Developed a Plotly web dashboard for live light curve simulations, dynamic transit generation, and real-time model comparisons.
- Finassist | Personal Finance PWA - [GitHub](#) | [Deployment](#)**
  - Engineered a Google Gemini LLM agent with 8 custom tool-use/function-calling schemas and a RAG retrieval system for zero-friction conversational transaction logging and financial onboarding with strict advice guardrails.
  - Architected a FastAPI backend across 14 modular REST routers and 13 relational tables, implementing automatic multi-account balance reconciliation, recurring bill engines, and 2.5x threshold spending anomaly detection.
  - Built a mobile-first React 19 PWA with Service Worker offline caching, background transaction queues, and dynamic Recharts data visualisations for 50/30/20 budgeting and net worth tracking.

### SKILLS & LANGUAGES

- Languages & Core CS:** Python, Java, C, SQL, Data Structures & Algorithms (DSA), OOP
- AI/ML Frameworks:** TensorFlow, Scikit-Learn, MLflow, Pandas, NumPy, RAG, LLM
- Web & Cloud:** Flask, Streamlit, React.JS, MS Azure (AI-900), Oracle Cloud (OCI)
- Tools:** Git/GitHub, Tableau, PowerBI, Ollama, Hermes
- Leadership Roles:** Coordinator - Texperia 2026, Codexia, Cybersecurity Workshop | Executive Member - IEEE, Gennuera

### ACHIEVEMENTS & CERTIFICATIONS

- Certifications:** [Microsoft Azure AI Fundamentals \(AI-900\)](#), [Oracle Certified AI Foundations Associate](#), [NPTEL Programming with Generative AI](#), [AWS Job Simulation](#), [NIT-T GenAI Workshop](#)
- Solved 200+ DSA Problems on LeetCode - [Link](#)
- 2x Winner** @ SPARK Robotics Competition, ACE College of Engineering, Trivandrum
- Winner of Codexia and CodeSprint Hackathons** @ SNS Institutions
- Presenter, Intl. Conference** - Sathyabama University (in collaboration with Univ. of Nottingham & ITS)